Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 39)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

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0 1 . 2 Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

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0 1 . 3 Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

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0 2 . 1 Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

[1 mark]

A Environmental DNA (eDNA) analysis ☐

B Radio tracking ☐

C Lincoln index (mark-release-recapture) ☐

D Leg rings ☐

0 2 . 2 Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

[2 marks]

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0 2 . 3 Explain how DNA databases could help to control the illegal trade in elephant ivory.

[2 marks]

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0 3 . 1 Explain **three** features of natural systems that make them sustainable.

[3 marks]

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0 3 . 2 Explain why the human use of mineral resources makes their sustainable exploitation much harder than the use of materials in a natural cycle.

[2 marks]

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0 3 . 3 Explain **three** principles of the circular economy that should be applied when a new product is designed.

[3 marks]

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0 3 . 4 Explain why an activity that is sustainable for a small population may be unsustainable for a large one. Use **one** example.

[3 marks]

0 3 . 5 Explain why urban populations are less likely than rural communities to notice that a resource is being over-exploited, and give **two** consequences of this.

[4 marks]

15

0 4 . 1 Explain why electronic meters must be calibrated before use and describe how a pH meter is calibrated.

[3 marks]

0 4 . 2 Describe how wind velocity measurements with an anemometer should be standardised.

[3 marks]

0 4 . 3 Describe how a whirling hygrometer is used to measure humidity.

[2 marks]

0 4 . 4 Describe **two** methods of measuring nitrate concentration in water.

[2 marks]

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0 5 Table 1 shows the number of bird species and butterfly species recorded in three woodlands in the same region.

Table 1

Woodland	Management	Bird species	Butterfly species
A	Ancient oak wood, coppiced on rotation with clearings	42	23
B	Ancient oak wood, unmanaged for 60 years, dense canopy	28	9
C	40-year-old conifer plantation	14	3

0 5 . 1 Use Table 1 and your knowledge to explain the differences in biodiversity between the three woodlands.

[5 marks]

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0 5 . 2 Describe **three** conservation efforts used to protect UK woodlands.

[3 marks]

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0 5 . 3 Explain why a high biodiversity increases the ecological stability of a woodland.

[2 marks]

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0 6 Table 1 shows some designated protected areas in the UK.

0 6 . 1 Complete Table 1 to show the legislation under which each area is designated and its key feature.

[4 marks]

Table 1

Designation	Designated under	Key feature	
Site of Special Scientific Interest (SSSI)			
Special Protection Area (SPA)			
Ramsar site			
Marine Conservation Zone (MCZ)			

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0 6 . 2 Explain how designating a protected area protects habitats and species.

[3 marks]

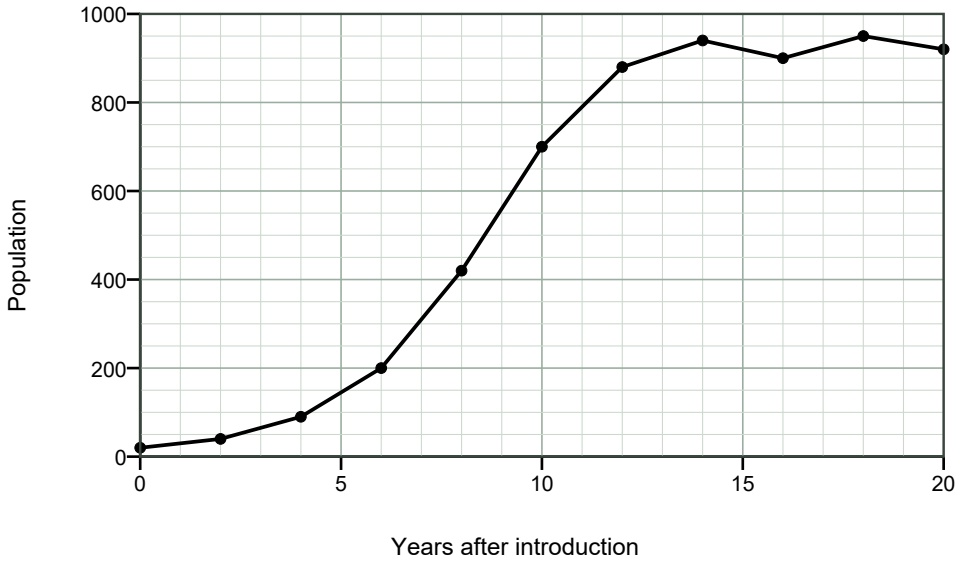
0 6 . 3 Explain why voluntary agri-environmental schemes are needed in addition to legislation.

[3 marks]

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0 7 Figure 1 shows the population of red deer on an island after they were introduced in year 0. No predators are present.

Figure 1



0 7 . 1 Name the shape of the growth curve and explain the changes in growth rate shown.

[4 marks]

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0 7 . 2 Explain the difference between density-dependent and density-independent factors, with an example of each.

[2 marks]

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0 7 . 3 Explain why the deer population on the island may need to be culled even though it appears stable.

[2 marks]

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0 7 . 4 Explain how predator and prey populations regulate each other.

[2 marks]

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0 8 . 1 Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

A Close planting of a single non-indigenous species ☐

B Mixed indigenous species with a mixed age structure ☐

C Clear-felling on a 40-year rotation ☐

D Removal of dead wood ☐

0 8 . 2 Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

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0 9 A new lowland nature reserve is being created on former farmland next to an existing ancient woodland.

0 9 . 1 Explain why the **area** and the **shape** of the reserve both matter in its design.

[3 marks]

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0 9 . 2 Explain the value of a biological corridor linking the new reserve to the ancient woodland.

[2 marks]

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0 9 . 3 Explain how each of the following abiotic factors could be managed to increase the biodiversity of the reserve.

[4 marks]

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Water levels .....

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Soil nutrient levels .....

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Topography .....

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Light levels .....

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0 9 . 4 Much of the reserve will be kept as a **plagioclimax**. Explain what this means and how grazing is used to maintain one.

[3 marks]

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0 9 . 5 Explain why the removal or culling of some species may be necessary on the reserve.

[3 marks]

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1 0 . 1 Evaluate the extent to which the principles of the circular economy can be applied to **three** of the following: water supplies, mineral supplies, waste management, pollution control, energy supplies, food production.

[9 marks]

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Extra space

1 0 . 2 State what is meant by **industrial symbiosis**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

OR

1 1 . 2 Discuss how human activities influence biodiversity, using examples from a range of different contexts.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS